Appendix B – Calliope Assessed

The research has assessed Calliope primarily against Scandroid. This section compares Calliope to expert scansion and the systems of Plamondon, Raabe, Fabb and Groves to determine how effective it is against more recent sophisticated theories. It also assesses Calliope against a Scandroid prototype, revealing some of Hartman's assumptions about scansion.

B1 Expert scansion

B1.1 Identification of Meter in Difficult Lines

Calliope is able to identify the meter in lines which are classified as being difficult or impossible to scan successfully. For example, Steele (1999) comments:

> "were we to encounter anywhere but in The Song of Hiawatha ('Introduction',34),
> The blue heron, the Shuh-shuh-gah
> it is unlikely that we would emphasize the two definite articles at the expense of 'blue' and the first syllable of 'Shuh-shuh-gah.' But that is what Longfellow wishes us to do, since he is writing in trochaic tetrameter:
> The(s) blue(w) her(s)on(w), the(s) Shuh(w)-shuh(s)-gah(w)
> … why can we not consider Longfellow's line about the heron as being trochaic with an iambic substitution in the first foot and an iambic substitution after a mid-line pause?
> The(w) blue(s) her(s)on(w), the(w) Shuh(s)-shuh(s)-gah(w)
> To this question, I would respond that, unless we stress the first syllable of Longfellow's line we lose the meter. It dissolves."

However, from a phonological perspective, the line shows only a very minor deviation from regular trochaic tetrameter. The two syllables ending the intonational unit and line ending are both regular trochees, only the first Shuh of polysyllabic word Shuh-shuh-gah indicates a iambic pattern: {[The(w) blue(s)] [her(s)on(w)]}>, [the(w) Shuh(m)-shuh(s)-gah(w)]}. Calliope has no difficulty in identifying a trochaic rhythm without adjusting the linguistic stress pattern.

B1.2 Assessed against Human Expert Scansion and Scandroid

The following analysis is of a poem by E. Woolley, given online (Woolley, 2006). The poem is scanned by four contributors (scottish jenny, brian w, Michael Wren, and boromir). I have also scanned it using Stallings's principles (detailed in 4.5.1), and with Scandroid and Calliope.

> Extraction Woes, by E. Woolley
> "I wonder why we call them "wisdom" teeth.
> Vestigial structures crowd, impact, occlude,
> make mastication painful; often food
> like steak becomes a hazard to the mouth.
> Oh, bloody day! Extractions take their toll;
> You'll slurp potato soup. No straws! Suck gauze.

Relief sought in a Hydrocodone haze,
and packs of Sonic ice fall short. You'll still
be numb from cheek to chin; cry out for Mom
who'll barely recognize your puffy face.
Soft tissue swells, dry sockets slow the pace
of healing -- weeks of agony to come.
I had the surgeon put mine in a jar
so I could see them when I start to whine
about the boss' temper or the swine
who grabs my ass each time I pass his chair.
I ask myself how does a long lunch line
compare to plastic props that wretched my jaw?
Do traffic-jams seem half as irksome now
when oozing orifice memories remain?"

Line	scottish jenny	Brian w	Michael Wren	boromir	Stallings (expert)
same	9	15	16	17	20
%	**43%**	**75%**	**80%**	**85%**	**100%**
error	8.5	1	1	1.5	0
%	**43%**	**5%**	**5%**	**8%**	
Total	**3%**	**70%**	**75%**	**78%**	**100%**
1	wswswswsws	wswswswsws	wswswswsws	wswswswsws	wswswswsws
2	wswswsswws	wswswswsws	wswswswsws	wswswswsws	wswswswsws
3	sswwwswsws	wswswswsws	wswswswsws	wswswswsws	wswswswsws
4	wswswswwws	wswswswsws	wswswswsws	wswswswsws	wswswswsws
5	sswswswsws	sswswswsws	wswswswsws	wswswswsws	wswswswsws
6	wswswswsws	wswswswsws	wswswswsss	sswswswsws	wswswswsws
7	wswswwswws	wswswwswss	wsswwwssws	wsswwswsws	wsswwswsws
8	wswswswsws	wswswswsws	wswswswsws	wswswswsws	wswswswsws
9	wswswswsws	wswswswsws	wswswswsws	wswswswsws	wswswswsws
10	wswswwwsws	wswswswsws	wswswswsws	wswswswsws	wswswswsws
11	wswswswsws	wswswswsws	wswswswsws	wswswswsws	wswswswsws
12	wswswswwws	wswswswsws	wswswswsws	wswswswwws	wswswswsws
13	wswswsswws	wswswswsws	wswswsswws	wswswsswws	wswswsswws
14	wswswwwsws	wswswswsws	wswswswsws	Wswswswsws	wswswswsws
15	wswsswwss	wswswswsws	wswswswsws	Wswswswsws	wswswswsws
16	wswswswsws	wswswswsws	wswswswsws	Wswswswsws	wswswswsws
17	wswswswsws	wswswswssw	wswswswsss	Wswswswsws	wswswswsws
19	wswswswsws	wswswswsws	wswswswsws	Wswswswsws	wswswswsws
20	wswsswwsws	wswswswsws	wswswswsws	Wswswswsws	wswswswsws
21	wswswwswwws	wswswwswsws	wswswsswsws	Wswswwswsws	wswswsswws

Table 62: scansion of Extraction Woes by human contributors (Woolley, 2006)

Line	Scandroid	Calliope	Stallings (expert)
Same	4	15	20
%	**19%**	**75%**	**100%**
Error	5.5	2	0
%	**28%**	**5%**	
Total	**-8%**	**70%**	**100%**
1	Wswswswsws	wswswswsws	wswswswsws
2	Swwswsswsw	wswswswsws	Swwswsswsw
3	Swwswswsws	swwswswsws	swwswswsws
4	Wswswswsws	wswswswsws	wswswswsws
5	Sswswswsws	sswswswsws	Sswswswsws
6	Ssswwsssss	wswswswsws	Ssswwsssss
7	Wsswwwswws	wsswwwswws	wsswwwswws
8	Wswswsssss	wswsssswsws	Wswswsssss
9	Wswswssws	wswswswsws	Wswswssws
10	Sswswswsws	wswswswsws	Sswswswsws
11	Sswssswsws	wswswswsws	Sswssswsws
12	Wswswwswws	wswswswsws	wswswwswws
13	Wswswwsswws	wswswsswws	wswswwsswws
14	Wswswswsws	wswswswsws	wswswswsws
15	Wswssswsws	wswswswsws	Wswssswsws
16	Wswssswsws	wswswswsws	Wswssswsws
17	Wswswswsss	wswsswwsws	Wswswswsss
19	Wswswswsws	wswswswsws	Wswswswsws
20	Wswssswsws	wswswswsws	Wswssswsws
21	Wswswswsws	wswswsswsws	Wswswswsws

Table 63 scansion of Extraction Woes by Scandroid and Calliope

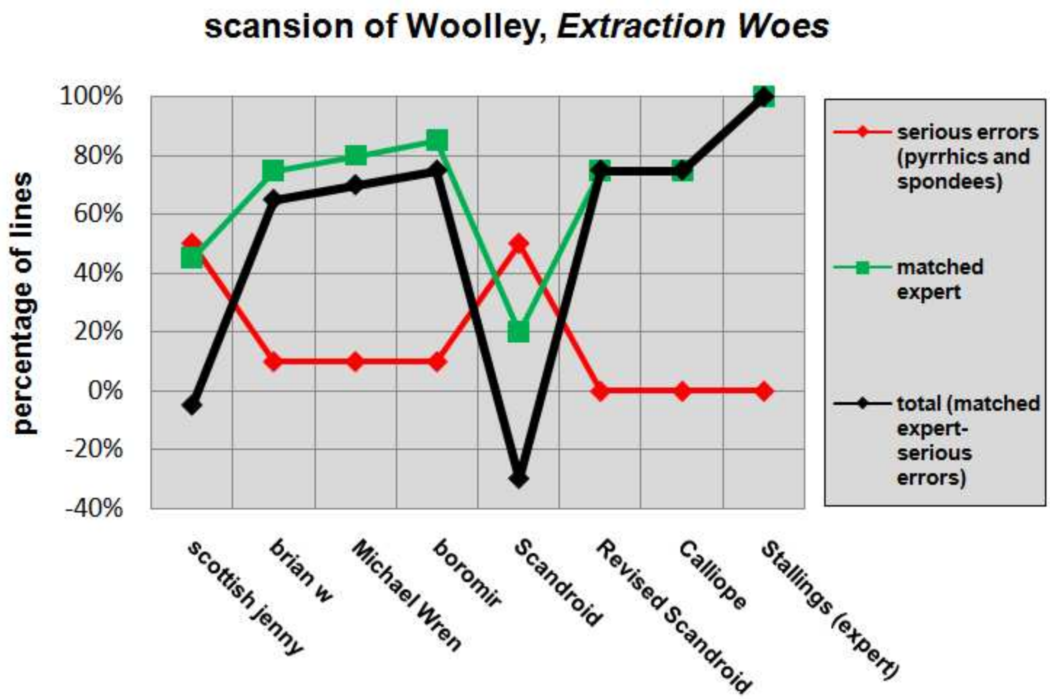

Figure 11: scansions of Woolley, Extraction Woes

From the scansion of "Extraction Woes":

1. Scandroid does far worse than the worst contributor
2. Calliope matches the value of one of the better contributors. However, two better scansions were contributed, although they were not much better.
3. The data replicates the results from the scansion of Pope's lines described above (4.5.2)

B1.3 Assessed against Unmodified Expert Scansion

The results of assessing regularised expert scansion against Scandroid and Phonological Scansion theory's scansion of 372 verses, categorised by complexity, are given in 4.2.2. The scansion was modified to screen out the subjectivity of expert scansions described in 2.1.3. In particular, the following:

1. A high percentage of expert scansions do not pick up on scansions like 'Shall(s) I(w) com(w)pare(s)' where there is a deviation from the meter in a phonologically insignificant part of the line (category 2). Instead experts produce the incorrect pattern 'Shall(w) I(s) com(w)pare(s)'. This is not so surprising, since Hayes and Kaun (1996) predict that it is precisely at these points that deviations from the standard meter are least likely to be detected.
2. Some expert scansions also permit sequences of strong syllables where one or more are subordinated to a main stress, although a greater number of experts prefer to subordinate one strong stress where possible (Baker et al, 1996). I have modified the expert scansion by replacing subordinate strong stresses with weak metrical stresses. This allows contiguous strong stresses where the stresses are not co-dependent.

Tests against unmodified scansions are given below as a comparison:

Category	1	2	3	4	5	6	Grand Total
Scandroid Match	77	3	2	19	14	1	116
Phonological Scansion Match	172	6	4	40	17	0	239
Count	215	20	4	83	49	1	372
% Scandroid	**35.8%**	**15.0%**	**50.0%**	**22.9%**	**28.6%**	**100.0%**	**31.2%**
% Phonological Scansion	**80.0%**	**30.0%**	**100.0%**	**48.2%**	**34.7%**	**0.0%**	**64.2%**
Target	80	80	80	60	60	71	72
Scandroid confidence	6%	16%	49%	9%	13%	0%	5%
Phonological Scansion confidence	5%	20%	0%	11%	13%	0%	5%

Table 64: assessment of Scandroid and Phonological Scansion against expert scansion by Scansion Complexity Categories

Assessed against unmodified expert scansions, Phonological Scansion performs better than, or as good as, Scandroid in all but one category. However, it falls far short of meeting the targets. I have accepted the modified expert scansion data as more accurate for the following reasons:

1. The modified scansion is consistent with the agreed principles of a representative sample of prosodists (Baker, 1996)

2. The data produced by modified expert scansion shows the expected decline in accuracy across Complexity Categories – the unmodified scansion shows an aberrant decrease in accuracy in Category 2 verses.
3. Additionally, the main difference between the modified and unmodified scansion data occurs in Category 2 verses, where there are variations in phonologically insignificant syllables. In other words, most people will not hear any significant difference between modified and unmodified scansions, but they do produce a disproportionately great effect on the accuracy of all the procedures being tested. It seems reasonable to regularise scansion in this category, giving the regular and most supported scansion the benefit of the doubt.

B2 Hartman's Scandroid prototype

B2.1 Introduction

Hartman (1996) describes a prototype of Scandroid using the scansion of Shakespeare's Sonnet 116 as a demonstration of its capabilities. I have compared this with Scandroid and Calliope's scansion of the same sonnet, in order to reveal some of Hartman's assumptions about appropriate scansions.

> "Let me not to the marriage of true minds
> Admit impediments. Love is not love
> Which alters when it alteration finds,
> Or bends with the remover to remove:
> O no! it is an ever-fixed mark
> That looks on tempests and is never shaken;
> It is the star to every wandering bark,
> Whose worth's unknown, although his height be taken.
> Love's not Time's fool, though rosy lips and cheeks
> Within his bending sickle's compass come:
> Love alters not with his brief hours and weeks,
> But bears it out even to the edge of doom.
> If this be error and upon me proved,
> I never writ, nor no man ever loved."

Hartman (1996) comments:

> "there's nothing in the computer's scansion that I'd mark wrong in a student paper, though there are lines that I would scan differently. (I read the fourth line 'Or bends with the remover to remove')"

B2.2 Analysis

Line	Scandroid prototype (1996)	Scandroid (2005)	Calliope & Revised Scandroid	Stallings (expert)	Hibbison
Same	6	4	8	14	6
%	**43%**	**29%**	**57%**	**100%**	**43%**
error	4	5	0	0	0
%	**29%**	**36%**	**0%**	**0%**	**0%**
Total	**7%**	**-9%**	**57%**	**100%**	**43%**
1	Swswwswwss	swswwswwss	swswwswsws	swswwswsws	wswswswsws
2	Wswswsswss	Wswswwssss	wswswsswws	wswswsswss	wswswswsws
3	Wswswswsws	wswswswsws	wswswswsws	wswswswsws	wswswswsws
4	Wswswswsws	wswswswsws	wsswwswsws	wsswwswsws	wswswswsws
5	Sswswswsws	sswswswss	wswswswsws	ssswwswsws	wswswswsws
6	Wswswswswsw	wswswswswsw	wswswswswsw	wswswswswsw	wswswswswsw
7	Wswswswswws	wswswswswws	wswswswsws	swwswswsws	wswswswsws
8	Wswswswswsw	wswswswsssw	Wswswswswsw	wswswswswsw	wswswswswsw
9	Sssswswsws	sssswswsws	Swwswswsws	sswswswsws	wswswswsws
10	Wswswswsws	wswswswsws	Wswswswsws	wswswswsws	wswswswsws
11	Sswswwssws	sswswwssws	Swwsswwsws	sswswswsws	wswswswsws
12	Wswsswwwsws	wswsswwwsws	Wswswswsws	wswsswwsws	wswsswwsws
13	Wswswswsws	wswswswsws	Wswswswsws	wswswswsws	wswswswsws
14	Wswswsssws	wswswsssws	Wswswsswws	wswswsswws	wswswswsws

Table 65: scansions of Sonnet 116

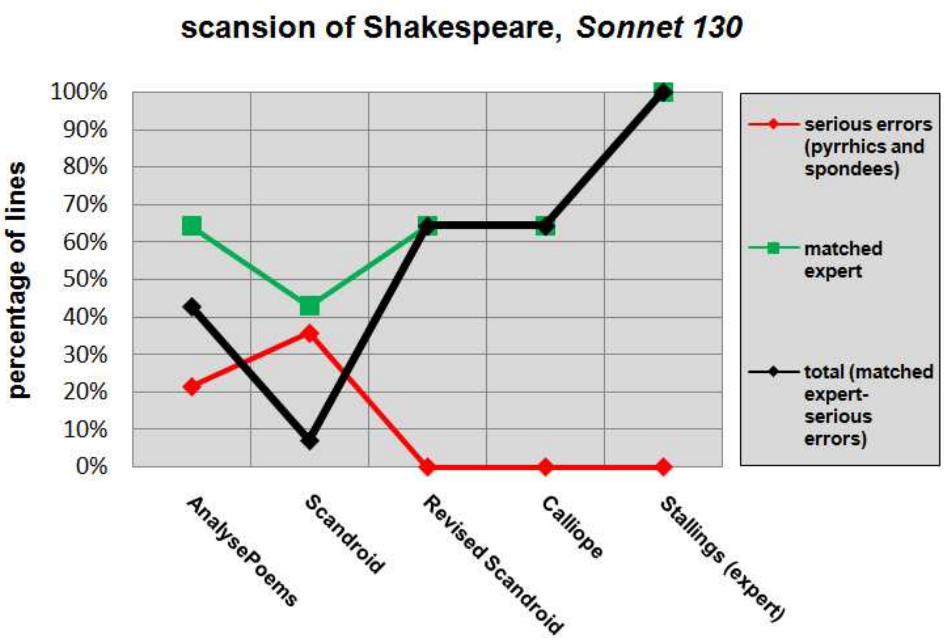

Figure 12: scansions of Shakespeare, Sonnet 116

The results show:

1. The Scandroid prototype matches more lines than the Scandroid program
   a. The prototype makes fewer errors than the program because the program assigns strong stress to the verb "to be" more frequently, sometimes ignoring a scansion which agrees with Stallings
   b. The prototype is more accurate than the program, but both frequently make (what Stallings considers) serious errors in their scansions. If Hartman allows these in undergraduates' work, then he is allowing errors which are avoided by the majority of prosodists.
2. Calliope performs better than a literary expert (Hibbison, 2005) when measured against Stallings's scansion. Hibbison allows relatively stressless syllables to be stressed to meet iambic norms - for example "not" (line 1), and "the" (line 4) - whereas Calliope avoids this.
3. Calliope differs from Stallings's scansion partly because it demotes spondees (ss) too frequently (lines 2, 5, 9, 11 and 12). For some prosodists (Baker, 1996) this practice is preferable to allowing spondees, and would probably be acceptable to Stallings. The other deviations arise from an acceptable difference of emphasis on words – Stallings stresses "it" in line 7, Calliope "is". Whereas Scandroid produces serious scansion errors, Calliope's errors are minor, if they are even errors at all.
4. Scandroid using the revised stresses produces an identical scansion to Calliope.

B2.3 Summary

This test again shows that Calliope outperforms Scandroid and approximates a very good scansion. It also shows that Hartman allows Scandroid to produce what at least one prosodist considers to be serious scansion errors. This has limited the effectiveness of the program.

B3 Plamondon's AnalysePoems

Calliope is compared to AnalysePoems (see 2.6.5) using two criteria:

1. Accuracy in identifying the meter of poems
2. Accuracy in identifying the rhythm of individual feet

B3.1 Identification of Meter

The table below compares the Calliope application with AnalysePoems in the identification of meter. The lower the cumulative difference from expert rankings the better the application approximates expert scansion.

Poem	AnalysePoems	Diff	Rank	Meter	Inversion	rank	Calliope	%	Rank
To Anthea, Herrick	87	16	1	Iambic	97	1		100	2
A daughter of Eve, Rossetti	87	21	2	Iambic	92	3	60/63	95.2	4
Amy Margaret, Allingham	86	7	3	mixed iambic / trochaic	95	2	90/90	100	2
Ulysses, Tennyson	84	21	4	Iambic	91.9	4	96/104	92.3	6
To the memory of Mr Oldham, Dryden	84	23	5	Iambic	88.3	5	96/99	97	3
Sonnet 130, Shakespeare	83	39		Iambic	90				
My Last Duchess, Browning	83	27		Iambic	80				
When I consider how my light, Milton	79	19	6	Iambic	80	7	46/55	83.6	7
Ode to a Nightingale, Keats	73	22	7	Iambic	88	6	78/82	95.1	5
Out, Out, Frost	73	25		free verse					
Dover Beach, Arnold	69	20	8	mostly iambic	77	8	54/66	81.8	8
Cumulative difference from expert rankings			4			0			10

Table 66: An assessment of AnalysePoems (yellow data) against Calliope (blue data) compared to expert scansions (white data).

B3.1.1 Regularity of line rhythm

The 'Inversion' column gives the percentage of feet which do not fit the overall metrical pattern. This is the benchmark against which to assess AnalysePoems and the phonological scansion. The poems are ranked from the highest percentage of inversions to the lowest (white rank column).

AnalysePoems's scansion ranks the poems in the same order as the inversion information (yellow rank column).

Calliope does not. The data in the column 'Calliope' gives the weighting of patterns matched in different phonological units (where Clitic Phrases have less weight in the total than Intonational Units). The ranking (in the blue rank column) does not match the inversion rankings. This may not be surprising since it is meant to approximate the perception of the listener, who will ignore some inversions as less relevant. It is, though, difficult to measure whether this ranking corresponds to the actual perception of listeners – that is, do the poems of Keats and Dryden sound more rhythmical than those of Milton and Rossetti?

B3.1.2 Discrimination of meters

However, Calliope is able to identify that there is a distinct difference in the meters of Out, Out and Ode to a Nightingale. AnalysePoems classifies both as duple meter with a confidence level of 73%, and a difference between first and last assessments of around 23%. However, Out, Out is free verse, according to both experts and the Calliope. Calliope also makes the type of duple meter explicit, where AnalysePoems probably has the data, but does not reveal it.

B3.1.3 Conclusion

AnalysePoems is better at identifying the regularity of meter. However, Calliope is better at identifying meters (such as iambic or free verse), and perhaps at distinguishing between subtypes of meter (such as iambic pentameter as opposed to iambic tetrameter).

B3.2 Identification of Feet

Plamondon (2006a) gives a detailed scansion of Shakespeare's Sonnet 130. I have compared this to Calliope's scansion and Raffel's (1992) expert scansion of the same sonnet in order to determine which matches the expert scansion of individual feet more accurately. A summary of the differences is given on p. 185, and conclusions drawn.

B3.2.1 Analyse*Poems's* Scansion

my(1) MISTRESS'(2) –(0) eyes(2) are(1) no(2)thing(0) LIKE(2) the(0) sun(2)
coral(0) –(0) is(1) far(1) more(1) red(2) than(0)her(0) lips'(2) red(2)
if(0) snow(2) be(0) white(2) why(1) THEN(2)her(0) breasts(2) are(1) dun(2)
if(0) hairs(2) be(0) WIRES(2) black(2)WIRES(2) grow(2) on(0) her(0) head(2)//
i(1) HAVE(2) seen(1) ro(2) ses(0)DAMASK'D(2) –(0) red(2) and(1) white(2)
but(1) NO(2) such(1) ro(2) ses(0) see(2) i(1)IN(2) her(0) cheeks(2)
AND(2) in(0) some(1) PERFUMES(2) –(0) IS(2)there(1) MORE(2) de(0) light(2)
than(0) in(0) the(0) breath(2) that(1) FROM(2)my(1) MISTRESS(2) –(0) reeks(2)//
i(1) love(2) to(0) hear(2) her(0) speak(2) yet(1)WELL(2) i(1) know(2)
that(1) mu(2) sic(0) HATH(2) a(0) FAR(2)more(1) plea(2) sing(0) sound(2)
i(1) grant(2) i(1) ne(2) ver(0) SAW(2) a(0)GODDESS(2) –(0) GO(2)
my(1) MISTRESS(2) –(0) WHEN(2) she(1)walks(2) treads(2) on(0) the(0) ground(2)//
and(1) yet(1) by(0) heav'n(2) i(1) think(2)my(1) love(2) as(1) RARE(2)
as(1) a(2) ny(0) SHE(2) belied(0) –(2) with(0)false(2) com(0) pare(2)

ns|ws|ws|ws|ws
ww|nn|ns|ww|ss
ws|ws|ns|ws|ws
ws|ws|ss|sw|ws

B3.2.2 Phonological Scansion from Calliope

{[MY(m) mis(s)tress'(w)] [eyes(s)]} {[are(n)]} {[noth(s)ing(w)] [like(s) the(w) sun(s)]}
Xsw]s}n]sw]xws} = (10) (Iambic = final PP, 3CP, 1PP) = Iambic) -> ms/ws/ns/ws/ws

[Co(s)ral(w)]} [is(n) far(m)] [more(n)] [red(s)]} [than(n) her(w) lips'(s)] [red(m)]}>:
sw}nm]n]s}xxs]m> = (10) (Iambic=1CP, 1PP, Trochaic=final IU, 1PP) =Trochaic -> sw/nm/ns/nw/sm
wn->W, sm->s-m = sW}m]n]s}xxs]-m> = (10) (Iambic=final IU, 1CP, Trochaic=2PP, 1CP) =Trochaic ->sW/mn/sn/ws/-m

[If(n) snow(s)] [be(n) white(s)]}>, [why(s)] [then(n) her(w) breasts(s)]} [are(m) dun(s)]};
=xs]ns>s]xws}ms} =(10) (Iambic=final PP, 1PP, 1IU, 1CP, Trochaic=1CP) =iambic ->ns/ns/sn/ws/ms

{[If(n) hairs(s)] [be(n)]} {[wires(s)]}, {[black(s)] [wires(m)]} {[grow(s)]} {[on(w) her(w) head(s)]}
Xsn}s}>sm}s}xws}= (10) (Iambic=finalPP, 1PP, 1IU , Trochaic=2PP) iambic?
(stress clash ssms> resolve (ww>w, s>s> s>-s = xsm}s}-sm}s}Ws}) (Iambic = final PP, 3PP(ms,ms, -s) 1CP) Iambic -> ns/ns/-s/ms/Ws
(no stress clash > (s>s) (Iambic final PP, 1PP, 1IU (ns), Trochaic = 1PP(sm), 1PP) = Iambic ->ns/ns/sm/sw/ws

ms|ws|ns|ws|ws
sw|mn|sn|ws|-m
ns|ns|sn|ws|ms
ns|ns|sm|sw|ws
ms|ws|ns|ws|ws

B3.2.3 Scansion by Raffel (1992)

My(w) mis(s)tress'(w) eyes(s) are(w) no(s)thing(w) like(s) the(w) sun(s)
Co(s)ral(w) is(w) far(s) more(w) red(s) than(w) her(s) lips'(w) red(s):
If(w) snow(s) be(w) white(s), why(w) then(s) her(w) breasts(s) are(w) dun(s);
If(w) hairs(s) be(w) wires(s), black(w) wires(s) grow(s) on(w) her(w) head(s).

ws|ws|ws|ws|ws
sw|ws|ws|ws|ws
ws|ws|ws|ws|ws
ws|ws|ws|sw|ws

B3.2.4 Summary

AnalysePoems	differences	Calliope	differences	Raffel
ns/ws/ws/ws/ws	All match	ms/ws/ns/ws/ws	All match	ws|ws|ws|ws|ws
ww/nn/ns/ww/ss	3 pyrrhic feet + 1 spondaic foot (v. unusual), 1 foot matches	sW/mn/sn/ws/-m	3 feet match, elision (not unusual), missing syllable (unusual)	sw|ws|ws|ws|ws
ws/ws/ns/ws/ws	All match	ns/ns/sn/ws/ms	4 feet match	ws|ws|ws|ws|ws
ws/ws/ss/sw/ws	4 feet match, 1 spondaic foot (unusual)	ns/ns/sm/sw/ws	4 feet match, 1 trochee (v. common)	ws|ws|ws|sw|ws
ns/ws/ws/ws/ws	All match	ms/ws/ns/ws/ws	All match	ws|ws|ws|ws|ws

Table 67: comparison of the scansions of Sonnet 130 produced by AnalysePoems, Raffel and Calliope

AnalysePoems and Calliope both match about the same number of feet as Raffel's expert scansion. However, when trying to find the best scansion, AnalysePoems produces more unusual patterns (pyrrhics and spondees) than Calliope (which resolves these using elision and insertion of syllables). In fact, a line containing four pyrrhic (ww) or spondaic (ss) feet is unique in English poetry (so far as I can tell). Both AnalysePoems and Calliope cannot reference rhetorical or contrastive stress, and so are unable to determine the scansion of 'her' in "Co(s)ral(w) is(w) far(s) more(w) red(s) than(w) her(s) lips'(w) red(s)". However, Calliope's four-level stress system is able to identify the relative strengths of "is(w) far(s) more(w) red(s)" where AnalysePoems's three-level stress system produces "is(w) far(w) more(s) red(s)" .

B3.2.5 Conclusion

Although neither AnalysePoems's system nor Calliope can reference key pragmatic information, Calliope produces less unlikely scansions by using four levels of stress, and by employing elision and insertion.

B4 Raabe's Frost program

B4.1 Description

Raabe (1975) develops an automated scansion system, implemented in SNOBOL on an IBM 360/365, with a dictionary to assign three levels of stress to lines of poetry (including one ambiguous stress which can be resolved according to the values of contiguous stresses). He uses the system to scan Frost's entire corpus, primarily to verify expert assumptions by identifying statistical patterns in Frost's poetry (along the lines of Slavic Metrists like Tarlinskaja).

He himself identifies some improvements to the system (which he does not implement) including regularising the stress patterns by raising secondary stresses to primary stresses if surrounded by weak stresses, and lowering secondary stresses to weak stresses if surrounded by primary stresses. He is also aware that sequences of three weak stresses conform to the metrical pattern by promoting appropriate weak stresses to strong.

B4.2 Assessment

I have assessed Raabe's system's scansion of Frost's Desert Places against Calliope and Scandroid, compared to an expert scansion of the poem using the criteria identified for Stallings's assessments. The expert scansion consists of the linguistic stresses of each line fitted into disyllabic feet. I have assessed Raabe's original system as well as the system modified by the improvements he suggests which are detailed above. The results of the assessment are given in Table 68 below.

Desert Places
"Snow falling and night falling fast, oh, fast
In a field I looked into going past,
And the ground almost covered smooth in snow,
But a few weeds and stubble showing last.
The woods around it have it--it is theirs.
All animals are smothered in their lairs.
I am too absent-spirited to count;
The loneliness includes me unawares.
And lonely as it is that loneliness
Will be more lonely ere it will be less--
A blanker whiteness of benighted snow
With no expression, nothing to express.
They cannot scare me with their empty spaces
Between stars--on stars where no human race is.
I have it in me so much nearer home
To scare myself with my own desert places"

Lines	Expert	Scandroid	Raabe	Raabe Revised	Calliope
Same	16	5	1	7	14
%	100%	31%	6%	44%	87.5%
Error	0	8.5	10	4.5	0
%	0%	53%	63%	28%	0%
Total	**100%**	**-22%**	**-56%**	**16%**	**81%**
1	wswsswwsws	sswwsswsss	sswwsswsss	sswwsswsss	wswswswsws
2	swswswwsws	wwssssws ws	wwswwswsws	wwsswswsws	swswswwsws
3	wwsswswsws	wwsswswsws	wwsswswsws	wwsswswsws	swswwswsws
4	swwswswsws	wwsswswsws	wwsswswsws	wwsswswsws	swwswswsws
5	wswswswsws	wswswswsws	wswswwwwws	wswswswsws	wswswswsws
6	wswswswsws	swswwswsws	sswwwswwws	sswwwswsws	wswswswsws
7	swwswswsws	wsssws wsws	wwsswswwws	wwsswswsws	swwswswsws
8	wswswswsws	wwswswswsws	wswwwswsws	wswswswsws	wswswswsws
9	wswswswsws	wswwswswssw	wswwwwwsww	wswswswsws	wswswswsws
10	wswswswsws	wssswswsws	wwsswswwws	wwsswswsws	wswswswsws
11	wswswswsws	wswswswsws	wswswwwsws	wswswswsws	wswswswsws
12	wswswswsws	wswswswsws	wswswswwws	wswswswsws	wswswswsws
13	wswswswswsw	wswswswswsw	wswswwwswsw	wswswswswsw	wswswswswsw
14	wsswswwswsw	wsswssssw sw	wsswswsswsw	wsswswsswsw	wsswswwswsw
15	wswswswsws	wswswsssws	wwwwwwssws	wswswsssws	wswswswsws
16	wswsswwswsw	wswswwsswsw	wswswwsswsw	wswswwsswsw	wswsswwswsw

Table 68*: assessment of Raabe*'s system against Scandroid, Calliope expert scansion *of Frost's Desert Places*

Raabe's original system produces the worst scansion, even worse than Scandroid. It makes frequent significant errors, and only matches the expert scansion once. When it is revised with Raabe's suggested improvements it matches more expert scansion than Scandroid, and makes fewer significant errors – mainly because all the sequences of weak stresses are resolved. Calliope again produces the greatest number of matches to expert scansion, and no significant errors.

B4.3 Conclusion

Raabe's original system is the worse than Calliope and even worse than Scandroid – perhaps the worst system studied in this project. However, when it is revised with Raabe's suggestions it is much better than Scandroid (and perhaps as good as AnalysePoems). Nevertheless, Calliope is still significantly better.

B5 Fabb's Bracketted Grid Theory

B5.1 Christina Rossetti, Spring Quiet

Fabb uses Bracketted Grid Theory to determine the meter and rhythm of poems (see 2.5.2). Calliope and expert scansion agree with Bracketted Grid Theory in 4/5 of the test poems. However, Fabb's theory fails to identify the rhythm in Christina Rossetti's Spring Quiet.

Poem	Expert	Bracketted Grid Theory	Calliope
Barrett-Browning, Sonnets from the Portuguese 21	Iambic pentameter	Iambic pentameter	Iambic pentameter
Browning, The Lost Leader	Dactylic Trimeter	Dactylic Trimeter	Dactylic Trimeter
Charlotte Bronte, Diving	Anapaestic Tetrameter	Anapaestic Tetrameter	Anapaestic Tetrameter
Christina Rossetti, Spring Quiet	Trochaic Trimeter	**Irregular Dimeter**	Trochaic Trimeter
Christina Rossetti, Up-Hill	Iambic Pentameter	Iambic Pentameter	Iambic Pentameter

Table 69: comparison of Calliope with expert scansion and Bracketted Grid Theory

It fails in Spring Quiet because Fabb uses Stress Maxima to identify feet in "loose iambic" verse (and only in loose iambics). Where the procedure encounters a stress which is a stress maximum, it will assign that stress and the stress immediately before it to a foot, even if this causes the foot to contain more than two syllables. In effect, the foot structure is built around stress maxima.
For example in "arching high over" Fabb notes a stress maximum in 'high(w) o(s)ver(w)'. When the procedure comes to assign feet, instead of making o- (of 'over') the start of the foot, it also includes the entire stress maximum in the foot, '|high over|'.

<table>
<tr><th>Line</th><th>Fabb</th><th>Calliope</th><th>Expert</th></tr>
<tr><td>Gone were but the Winter</td><td>wws|[wsw]|</td><td>sm]|xxsw} (2 units = trochaic) = sm|nw|sw</td><td>sw|sw|sw</td></tr>
<tr><td>Come were but the Spring</td><td>wws|ws</td><td>sm]|xxs} (2 units= truncated trochaic) = sm|nw|s-</td><td>sw|sw|sw</td></tr>
<tr><td>I would go to a covert</td><td>wws|[wsw]| ('to a' = 1 stress)</td><td>xms]x[sw]} ('to a' = 1 stress) (2 units=trochaic) = sm|sn|w|sw</td><td>sw|sww|sw</td></tr>
<tr><td>Where the birds sing</td><td>ws|ws|</td><td>m}xs]xs} (stress clash of 'birds(s) sing(s)' resolved by inserting a stress) (3 units = truncated trochaic) = mw|s-|s-</td><td>sw|sw</td></tr>
<tr><td>Where in the whitethorn</td><td>ws|[wsw]|</td><td>m>xxxsw] (insert a stress between 'where' and 'in' because of the pause break) (2 units = trochaic) m-|nw|sw</td><td>sl|sw|sw</td></tr>
</table>

Table 70*: analysis of 'Spring Quiet' 1*-5 by Bracketted Grid Theory and Calliope

The table shows that Fabb's analysis fails for two reasons: firstly, his stress assignments are incorrect (perhaps motivated by the expected outcome). For example, Fabb stresses 'Gone were but' as wws with no stress on the two content words 'gone were'. Calliope stresses both 'gone' and 'but'. Secondly, Fabb's insistence that stress maxima can distort the meter is unnecessary. With a normal stress assignment, the lines are quite easily identified as trochaic trimeter by Calliope: there are no feet with an irregular rhythm, although two syllables need to be merged into one stress, and two unsounded stresses have to be inserted.

B5.2 Duple Meters with Ternary Rhythms

B5.2.1 Introduction

Malof (1964 and 1970), Kiparsky (1975), Attridge (1982), Hayes (1984), Groves (1998) and Fabb (2002) all note that there are some lines in which both duple and triple (or ternary) rhythms operate simultaneously. They attempt to identify how one or other of the rhythms can be shown to be dominant. Hayes excludes some of Attridge's scansions on linguistic grounds. Of the remaining lines, both his and Kiparsky's solution is to reference the rhythm of contiguous lines to fix the meter. Attridge, Groves and Fabb use their own systems to help choose a dominant rhythm, but in most cases are unsuccessful.

B5.2.2 Analysis

The table below (Table 71) details the comparison with Calliope – matches to expert scansion are coloured green:

<table>
<tr><th>Theory</th><th>Scansion</th><th>Comment</th></tr>
<tr><td colspan="3">a. Strangers to slander, and sworn foes to spite (John Pomfret, The Choice 3)</td></tr>
<tr><td colspan="3">Stran(s)gers(w) to(w) sla(s)nder(w) and(n) sworn(m) foes(s) to(w) spite(s)
(the expected rhythm is iambic pentameter to fit the poem's meter)</td></tr>
<tr><td>Iambic</td><td colspan="2">Strangers| to slan|der, and| sworn foes| to spite</td></tr>
<tr><td>Anapaestic</td><td colspan="2">Stran|gers to slan|der, and sworn| foes to spite</td></tr>
<tr><td>Expert</td><td>Iambic</td><td></td></tr>
<tr><td>Fabb</td><td>Strangers to slander and sworn foes
A.
)* *) * *) * *) * *) * *)
* * * * *
Sw/ws/ws/ws/ws
B.
)*) * * *) * * *) * * *)
* * * *
s/wws/wws/wws</td><td>Fabb notes that the line is in a iambic pentameter poem, and that it has one rhythm (A) which implies iambic pentameter. However, he comments "the arrangement of monosyllables also permits a different rhythmic explicature (B)…this …resembles…iambic…only weakly…deriving the conclusion that the line is anapaestic tetrameter (with a short initial foot)." He does not resolve the issue.</td></tr>
<tr><td>Calliope</td><td>=/sw]x/sw>xm]s}ws}
(Iambic=fPP,1PP,1IU,
Trochaic=1PW2,
Anapaestic=1CP,1IU,
No Triple=1CP,1PP)
iambic (7/8) or anapaestic(4/7)
=iambic (87.5%)</td><td>Calliope identifies a iambic rhythm in 3 out of 4 units (with a weighting of 7/8). It identifies an anapaestic rhythm in 2 out of 4 units (with a weighting of 4/7) – the final foot "foes to spite" produces a pattern of sws (an amphimacer) which cannot be construed as an anapaest. The overall rhythm is iambic with a consistency of 87.5%.</td></tr>
</table>

<table>
<tr><td colspan="3">b. When to the sessions of sweet silent thought (Shakespeare, Sonnet 30)</td></tr>
<tr><td colspan="3">When(s) to(n) the(w) ses(s)sions(w) of(w) sweet(s) si(m)lent(w) thought(s)
(the expected rhythm is iambic pentameter to fit the poem's meter)</td></tr>
<tr><td>Anapaestic</td><td colspan="2">When| to the ses|sions of sweet |silent thought|</td></tr>
<tr><td>Iambic</td><td colspan="2">When to| the sess|ions of| sweet sil|ent thought|</td></tr>
<tr><td>Expert</td><td colspan="2">Iambic Pentameter</td></tr>
<tr><td>Attridge</td><td>B ŏ B ŏ B ô B o B</td><td>Attridge inserts a beat between "sweet" and "silent", and merges "to the" and "-sions of" into single beats.
This creates a iambic line with anapaestic rhythms.
However, it is very unusual for Shakespeare to insert a beat, and unheard of for him to merge syllables into one beat (except using elision)</td></tr>
<tr><td>Groves</td><td>A-O_ o-A-o O_--a--A-o A
a.S w w S w w Ṣ w w S
b.Ś w w S w W s S w S</td><td>Groves tests both an anapaestic (a) and a iambic (b) reading of the line. Although his theory excludes pentameter readings of some tetrameter lines, it does not exclude either reading here. Neither reading has a weak lexical stress syntactically subordinated to a strong lexical stress, but placed in strong metrical position.</td></tr>
<tr><td>Calliope</td><td>=xxx/sw}xs/mw]s}
(Iambic=final PP,1PP,
Trochaic=1PW3,
Anapaestic=2PP,
No Triple=1CP, final PP)
Iambic (4/6=66.6%) or anapaestic (4/7=57.1%)
=iambic (66.6%)</td><td>Calliope does not insert or merge any syllables (keeping with Shakespeare's practice).
It determines that the iambic rhythm matches 2 of 3 units, with a weighting of 4/6 or 66.6% consistency.
It also determines that the anapaestic rhythm matches 2 of 4 units, with a weighting of 4/7 or 57.1% consistency. One unit has a rhythm of mws ("silent thought") and so is rejected – if it were accepted as anapaestic, the line would be anapaestic; the other unit is the initial "When".
The line is iambic, since iambic rhythm has a higher consistency.</td></tr>
</table>

<table>
<tr><td colspan="3">c. I look on the sky and the sea (Barrett-Browning, The Runaway Slave at Pilgrim's Point 7)</td></tr>
<tr><td colspan="3">I(n) look(s) on(n) the(w) sky(s) and(n) the(w) sea(s)
(the expected scansion is uncertain: previous lines are anapaestic or iambic; the number of strong stresses in each line is 4,4,3,3,4 and 3)</td></tr>
<tr><td>Anapaest ic</td><td colspan="2">I look| on the sky| and the sea|</td></tr>
<tr><td>Iambic</td><td colspan="2">I look| on the| sky and| the sea|</td></tr>
<tr><td>Expert</td><td colspan="2">Accentual</td></tr>
<tr><td>Fabb</td><td>I look on the sky and the sea
)* *) * *) * *) * *)
)* *] * (* *] * (* *]</td><td>The first scansion (with foot breaks marked by ')') is iambic, the second is anapaestic (with foot breaks marked by ']' because stress maxima are used to fix the meter). Fabb is unable to decide on a dominant rhythm in the line: he concludes "Neither option thus maintains the uniformity of the stanza. Again, we have disruption, exploiting a metrical ambiguity...which cannot be fully resolved…the metre is iambic-anapaestic"</td></tr>
<tr><td>Calliope</td><td>=ws}xws]xws}
=(8 syllables, 3 strong stresses)
(Iambic=1PP, finalPP,
Trochaic=1CP,
Anapaestic=fPP, 1CP
No Triple=1PP)
Iambic(4/5)
Or
Anapaestic(3/5)
= iambic (80%)
(but previous lines are anapaestic and iambic, with strong stresses of 4 433 43, indicating accentual Ballad Meter)</td><td>Calliope determines that 2 of 3 units have a iambic rhythm – with a weighting of 4/5. 2 of 3 units also have an anapaestic rhythm, but with a lower weighting of 3/5. Therefore, the line is iambic with a consistency of 80%.
Three previous lines have iambic rhythm, and three have anapaestic rhythm – so, the poem may be a unique mixed anapaestic/iambic meter. However, the strong stress pattern is 4 433 433 (including this line's 3 strong stresses), indicating that the poem is composed in the popular contemporary Ballad Meter of alternating 4 and 3 strong stresses per line.
Calliope determines the meter is neither iambic nor anapaestic but accentual.</td></tr>
</table>

Table 71: Ternary and Duple rhythms resolved

B5.2.3 Discussion

Calliope is able to determine the meter in lines which Fabb determine as irresolvably ambiguous (a and c). In one case (b), where Attridge resorts to processes which the poet would very probably not have used, Calliope reaches the same conclusion without modifying the line at all. In every case Calliope determines that there is a significant subordinate rhythm, but also identifies a dominant meter which fits with the expected pattern of the poem. In one case (c) it is able to determine that an accentual reading is the appropriate one, rejecting both anapaestic and iambic readings.

B5.2.4 Conclusion

Contrary to the assumption of Kiparsky and Hayes, the lines themselves in isolation are able to indicate the expected meter. In fact, in 30 test verses (14 verses from non-triple rhythm poems identified in the literature, and 16 verses from the anapaestic poems The Destruction of Sennacherib by Byron and Diving by Charlotte Bronte) Calliope identifies a dominant rhythm which is also the expected rhythm for the verse - whether anapaestic, iambic or accentual. The procedure appears to have identified a previously unnoticed scientific principle for consistently determining the meter in these cases. It also resolves the problem posed by Hayes: why poets writing iambic pentameter verse “scrupulously avoid” some formats of lines which have significant triple rhythms, but allow other formats with an even greater tendency for the same rhythms – from a phonological perspective, verses in the second category have dominant iambic rhythms.

B6 Groves’s Base Template Theory

Calliope produces very similar results to Groves’s Base-Template theory (2.5.1). However, there are two differences – the treatment of Stress Clash and the assignment of meter in some contexts. Both are discussed in detail below.

B6.1 Extended Stress Clash

Groves (1998) extends Hayes’s Stress Clash theory to make the line conform to metrical criteria.

B6.1.1 Hayes’s theory

Selkirk (1984) and Hayes (1984a) investigated cases where the stress in the first word of a compound is further back in the word than expected. They determined that where two contiguous syllables in different words or phrases contain stresses on a given phonological level not separated by at least one stress on the next lowest level on an intervening syllable, there is a stress clash. The clash is resolved by retracting the left stress back leftwards to the next most stressed syllable. An example is given in Table 72.

A hundred thirteen men	
x 1 0 x x x x x x x x x x x x x A hundred thirteen men	intial stress grid: clash between 1 and 0
x 1 x x 2 0 x x x x x x x x x x A hundred thirteen men	1 retracted to next available syllable stressed on the level below, clash between 2 and 0
x x x x 2 0 x x 3 x x x x x x x A hundred thirteen men	2 retracted to next available syllable stressed on the level below, no clash (the syllables 2 and 0 are separated by an intervening syllable stressed at the next lowest level, 3)
A(w) hun(s)dred(w) thir(s)teen(w) men(s)	result in terms of binary stresses

Table 72: Stress Clash in "a hundred thirteen men" from Hayes (1995)

More recently Hayes (1995) has identified that the stress can only be retracted to stressed syllables in the same phonological phrase.

B6.1.2 Stress Clash in Verse

The effect of stress clash on verse is less certain. Some think that it is matched to metrical stress, but can be reversed to fit the meter (Kiparsky, 1975 and Groves, 1998), or that different rules apply in different genres (Hanson, 2006). However, it seems more likely that compound dependents are assigned metrical positions determined by the primary stress in the (following) governing word (Koelb, 1979 and Youmans, 1983), and not by their own stress patterns. Whatever the process is, if two contiguous strong lexical stresses occur within a dependent noun, only the most metrically consistent one is heard by people reciting the verse. If stress clash cannot be resolved linguistically, then it is resolved by changing the word order – see Table 73.

Type		**Analysis**
A. In our well-found successes, to report (Shakespeare, Coriolanus 2.2)		
Stress Clash	…well(m)-found(s) suc(w)cess(s) -> …well(s)-found(m) suc(w)cess(s)	stress on 'found-' retracts to 'well'
Metrical (and heard) Stress	**..well(W)-found(S) suc(W)cess(S)**	'well' has weaker stress than 'found'
B. After a well-graced actor leaves the stage (Shakespeare, Richard II 5.2)		
Stress Clash	…well(m)-graced(s) ac(s)tor(w) -> …well(s)-graced(m) ac(s)tor(w)	stress on 'graced-' retracts to 'well'
Metrical (and heard) Stress	**..well(S)-graced(W) ac(S)tor(W)**	'well' has stronger stress than 'graced'
comment	Despite the same lexical stress pattern in A and B, B has a completely different metrical stress pattern, apparently due to the position of the primary stress in 'actor'. Moreover, A does not allow a lexically accurate and metrically acceptable trochee ('well(S)-found(W) suc(W)cess(S)') but prefers the compound to follow the stress pattern of 'success'.	
C. Buy terms divine in selling hours of dross; (Shakespeare, Sonnet 146)		
Stress Clash	…di(w)vine(s) terms(s) -> (cannot retract) -> **…terms(s) di(w)vine(s)**	Stress clash in divine(s) terms(s), but stress cannot retract on divine
comment	There is a stress clash between 'divine' and 'terms', but divine does not allow stress retraction. To avoid these sorts of clashes, divine is placed after its noun in most English poetry.	

Table 73: Stress Clash in verse

If Groves and Kiparsky are correct, according to data given by Fitzgerald (2003) the word order in lines would be changed to avoid the stress clash. Since very little word order change is apparent, this undermines Groves's theory. If Koelb and Youmans are correct, then the lexical stress of compound adjectives is irrelevant to their metrical position, though the lexical stress of compound nouns should correspond to the metrical positions – a theory supported by Table 73 and data in Hanson (2006). It seems that the second theory is more likely to be correct.
In any case, Phonological Scansion easily identifies the meter correctly without using Groves's stress analysis.

B6.2 Incorrect Assignment of Meter

Groves's theory identifies whether a line fits a particular meter; it can sometimes force an incorrect reading, where the line fits an alternate pattern but is identified with the pattern being tested for. One example, discussed below, is Wyatt's "They fle from me".

B6.2.1 Thomas *Wyatt's "They fle from me"*

"They fle from me that sometyme did me seke
with naked fote stalking in my chamber
I have sene theim gentill tame and meke
that nowe are wyld and do not remembre
that sometyme they put theimself in daunger
to take bred at my hand and nowe they raunge
besely seking with a continuell chaunge

Thancked be fortune it hath ben otherwise
twenty tymes better but ons in speciall
in thyn arraye after a pleasaunt gyse
When her lose gowne from her shoulders did fall
and she me caught in her armes long and small
therewithall swetely did me kysse
and softely said dere hert howe like you this

It was no dreme I lay brode waking
but all is torned thorough my gentilnes
into a straunge fasshion of forsaking
and I have leve to goo of her goodenes
and she also to use new fangilnes
but syns that I so kyndely ame served
I would fain knowe what she hath deserved"

Groves identifies the poem as loose form of iambic pentameter - in agreement with the assessments of Raffel (1992), Adams (1999) and others. If the poem is iambic, then it is very loose; so loose, in fact, that Wyatt's first editor, Tottel, felt the need to rewrite the poem to make it more regular. Daalder (1972) notes "Tottel's lines scan, or are meant to, where they do not in Wyatt".

However, Schwartz (1963) follows an argument of C.S. Lewis that C15 poems which appear to be irregular iambic pentameter are actually imitating the older tradition of accentual verse. He identifies a regular four-beat accentual pattern in Wyatt's poem:

> "we might suppose the poem to be written in irregular iambics. But if the lines are supposed to be pentameters (as most of them seem to be) we have difficulty accounting for seven lines…which will not yield more than four feet. And there are lines which sound very odd to an ear trained in iambic movements: lines seven, nine, twelve and seventeen…If we assume an accentual-syllabic norm for the poem, we get a slipshod and badly handled meter, and one which cannot produce the very real rhythmic power of the poem. If, however, we assume a four-stressed accentual norm, we get a firm and skilfully managed meter and we do not violate any rhetorical emphases...the scansion makes good rhythmical sense precisely at those points where violence is done by an iambic norm."

Calliope identifies this accentual pattern, rejecting a iambic or trochaic rhythm in some lines which Schwartz indicates do not fit the pentameter pattern. In these lines, Groves identifies a pentameter rhythm – see Table 74 – and rejects Schwartz's idea for three reasons: because it proposes a meter that is too simple and does not reference phonology in a rigorous way, but mainly because Wyatt tends towards using ten-syllable lines, implying a pentameter pattern. The following analysis is based in phonological theory, and supports Schwartz's conclusions:

Type	Analysis
Line	**With naked fote stalking in my chambre**
Groves	o------A--o A ^ A---o O o-----A---o w-----S w---S w-S w----S w-----S o
Calliope	with (w) na(s)ked(w)] fote(s)]} stal(s)king(w)] in(n) my(w) cham(s)bre(w)]} =x/sw]s}/sw]xx/sw} = (10,4) (Iambic=1PP, 1CP, Trochaic=1PP, 1CP) (/3/6) trochaic?? =(inserting weak stress ss-> s-s) x/sw]s}/-sw]xx/sw} = (11,4) (Iambic=fPP, 1PW2, 1PP, 1CP, Trochaic=) (6/6) iambic
Comments	Groves inserts a weak stress and creates a regular iambic rhythm. Calliope does the same thing, ignoring a possible accentual reading of the line.
Line	**That nowe are wyld and do not remember**
Groves	o------A o----Ao O O O o---A---o w------S w---S w—S [w-S] w---S o
Calliope	that(w) now(s)] are(w) wyld(s)]} and(w) do(s) not(m) re(w)mem(s)ber(w)]} xs]ws}x/smwsw}= (10,4) (Iambic=1PW5, 1PP, 1CP, Trochaic=fPP, 1PV5) (7/13) iambic???
Comments	Groves makes the word ‘wyld’ disyllabic (sw). Phonological Scansion is unable to identify a dominant rhythm for the line, and chooses an accentual reading.
Line	**That sometime they put theimself in daunger**
Groves	^O l A---b o---A O O o---A----o wS S---w w---S w----S w—S o
Calliope	{[That(w) /some(s)time(w)\] [they(n) put(s) /theim(w)self(s)] [in(n) /daun(s)ger(w)]} x/sw]ns\ws]xsw}= (10,4) (Iambic=2PW2, Trochaic=fPP, 2PW2) trochaic rhythm (4/6)
Comments	Groves inserts a weak stress at the start of the line Calliope identifies a weak trochaic rhythm, which may indicate that the line should be read as accentual.
Line	**Therewithall swetely did me kysse**
Groves	^A-----o O ^ A---o O o----A w-S w--S w-S w—S w---S
Calliope	{[There(s)with(w)all(n)] [swete(s)ly(w)] [did(m) me(n) kysse(s)]} /swn]/sw]/mns} = (8,4) (Iambic=1PW2, 1PW2, 1PP, Trochaic=1PW3) Iambic rhythm (4/6)
Comments	Groves inserts a weak stress at the start of the line, and after the first word. Calliope identifies a weak trochaic rhythm, which may indicate that the line should be read as accentual.

**Table 74: Groves*'s analysis of ambiguous lines in Wyatt's "They fle from me"*
compared to Phonological Scansion**

Since there is a variation of between 8 and 11 syllables in each line, but there are always four strong stresses, and since the meter of the lines varies and in many lines no rhythm is dominant, Calliope prefers to read the poem as four-beat accentual. Groves's theory allows him to read the poem as (an idiosyncratic) iambic pentameter.

B6.2.2 Conclusion

Calliope is able to assess a far wider range of scansion possibilities than Groves's analysis which has a tendency to produce false readings.